

One integer rule at every Node of a lattice: the click as the local,
whole measurement, the cloud as the body of nature, and the forms
of Einstein and Lorentz arrived at without them as inputs

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Abstract

We state one local rule in whole numbers, Rule3, applied at every Node of a cubic lattice each interval: a second-order step of a family's level from its two levels and its six neighbours' arrivals, with integer coefficients from the family's pair and the Node's paces, and one Euclidean division whose remainder stays at the Node. Every level on the lattice is written by that one act. A quantum moves whole when a remainder crosses the quantum's action, and that division is the click, the one measurement, local and whole: the record goes on after it, nothing ends anywhere, and no act at a distance is in the law. Exact on the lattice: the whole run returns bit for bit under the rule's own direction, a form is conserved, the count is conserved, and the momentum's flux is the wave's own stress. The families are bands of the rule. The body of nature is the cloud, the wide fixed point of the binding row, whose three masses (the well it sources, the count it resists with and the tail it falls with) are one to the order $(C/\Gamma)^2$; the compact pixel is collapsed matter. From the same coefficients, with a held level lowering the clock's pace once and the Link's pace once more, come the gravitational redshift, light's bending by twice the clock's share, the delay, the perihelion's 6π once the clock carries its second order, and Lorentz's factor at a body's own light speed; each form stands on the comparison side and none is an input. The first experiment run on the engine, the two slits, gives the fringes at the formula's spacing from the local click alone, beside a blind row written before the run, and two findings against nature written as they stand: a dilute packet reports nothing, and a bare detector in one gap changes nothing. The price of locality is stated plainly. The paper claims that the detectors' clicks represent reality and shows the match; it never claims that nature is so.

Keywords: integer lattice model; one local rule; the click as the measurement; locality; discrete relativity; pre-registered simulation

1 Introduction

The question. The law of this paper is one line of integer arithmetic applied at every Node of a cubic lattice at every interval. Nothing else acts: no field equation beside it, no law of motion for a body, no rate, no draw, no real number. The paper states that line, proves what is exact on the lattice, derives what follows from the line alone, names what is decided beside it, and sets the law's own numbers beside the forms of nature: the forms of Einstein, Lorentz, Newton, de Broglie and Young appear only on the comparison side, beside the law's expectation written before any run; none is an input. Every formula is taken from the law's document [1] and cited by the name of the

section it stands in; the engine that computes the rule is described by its own document [2], and the decisions in force by theirs [3].

The click's story, the spine of the paper. A click is the passage of one whole quantum from one Node to the next, at one Link, where a remainder kept at the Node crosses the quantum's action: the count's line's own division (Section 3.2). It is local, it is whole, and it ends nothing. The record, the wave on the lattice, goes on after it exactly as before; no wave collapses, no record ends anywhere, and no act at a distance is in the law. A detector is a Node declared in a file to report the rise of a family's count across its Nodes; it takes nothing and hands nothing over. Nature's detector, the paper says, is a body: a body reads the light's level at its Nodes by its own rotation sense into its paces, a drive at the light's frequency, resonant where it meets the body's rotation, and its write answers, the driven body writing light of its own that cancels the incoming wave behind it; where the wave is cancelled its current stops, the light's count that arrived at the body's Nodes does not leave, the count's line piles it there, and a Node told to report reports the rise: one photon at one resonant absorber clicks once, locally (Section 3.6). That is the whole story of measurement in this law, and its price is stated plainly in Section 3.7: a wave split among many absorbers gives each a fraction, so that at most one crosses or none, and no statistics are in it; a bare detector Node reports no spread quantum; and Bell's row is open. The paper calls these findings against nature and writes them as they stand.

The reading rule. Inside is inside the GameBoard, where no one measures; Outside is the game above the board, the detectors and their clicks. Only a detector's click is a measurement, compared with nature or pinned as an expectation. A number of the GameBoard's state (a level, a support, a total, a body's centre) is a GameBoard reading, a diagnostic, labelled so wherever it appears, never compared with nature and never a result on its own; a number whose kind is not named is not a result [1, readings and measurements]. The paper claims that the Outside represents reality and shows the match; it never claims that nature is so [3, inside and outside].

The marks. Every claim carries one of five marks: THEOREM, proved for every GameBoard; DERIVED, from theorems by argument; COMPUTED, a number of the law's own computed from its lines before any run; ASSUMPTION, a line the law holds by decision and names where it is used; HYPOTHESIS, a line outside the law under its own name, with what would decide it. A number of a run enters only through the documents' rows, labelled DETECTOR (a click, the measurement) or GAMEBOARD (a diagnostic).

The law in one line. At every Node, each interval, for every family [1, the line; the count's line; the primitives]:

Rule3:	$w a_{\text{next}} + r' = \sum_{a \in \{x, y, z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{ den } \Gamma^2, \quad R_a = 2 \text{ num } p_a^2,$ $S = 12 \text{ den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{ num } (p_x^2 + p_y^2 + p_z^2);$	(1)
the count's line:	$W_c c_{\text{next}} + r' = W_c c_{\text{now}} + \sum_j F_{ij} + r, \quad 0 \leq r' < W_c, \quad W_c = 3 \text{ den } T,$ $F_{ij} = \text{num} (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j);$	
the one write:	a level at a Node gains $(w q(\text{Node}) + r) \text{ div } E$, the remainder r kept there.	

In the first line a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval on; $\text{arr}_{\pm a}$ is the arrival through the Port $\pm a$, the neighbour's level; r and r' are the remainder kept at the Node before and after the step; Γ is the Node clock, the pace of an empty Node; $[\text{num}, \text{den}]$ is the family's pair, with $\cos \omega_0 = \text{num}/\text{den}$ its rotation per interval at

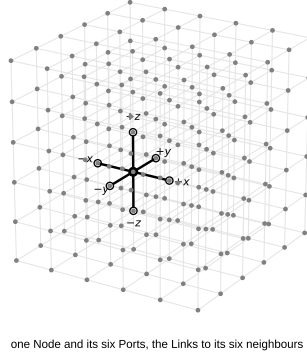


Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Fig. 2. A Node keeps each family's two levels and its remainders there, and nothing else. Drawn from the definitions; no run.

wave number zero; p_0 is the Node's clock pace and p_a its pace along the axis a , both the Node clock lowered by the levels the family reads there; and div is the floor division. The second line is the same rule for the family's quanta: c is the count, the number of quanta at the Node; T is the quantum's action, the one unit of every family; F_{ij} is the current into the Node i from its neighbour j over the interval, the bilinear form of the record's two levels at the two ends; and the division is the click, one quantum moving whole exactly when the remainder crosses the wall W_c . The third line is every write of a record into a held level: $q(\text{Node})$ is the writer's own quantity there, w the one weight with which the writer's family reads the written family, and E the written row's divisor. Everything else is a number in a file. The first line is a theorem's object (Section 2), the second the click (Section 3), the third the one write (Section 2.8).

The words. The physical location is a *Node*; its complete local information is its *NodeState*, the law's own numbers and nothing else. Nodes are joined by *Links* through six directional *Ports* per Node; the lattice of Nodes is the *GameBoard*; a local change is an *Event*, and there is one *LocalRule*, Rule3. A *Family* is one kind of physical value with its pair and what it holds; its *Record* is its levels over the GameBoard, two per Node with a remainder. A *Body* is the Nodes on which a family's count stands with the standing record the count binds, with no law of its own; a *Message* is a record the world lays and no body. A *Detector* is a Node declared in the file to report; its *Click*, the rise of a family's count across its Nodes, is the one measurement. An *Interval* is one step of the whole GameBoard [1, the objects][2, the words].

What this paper claims, and what it does not. Claimed: one explicitly stated integer rule whose reversibility, conserved form, count conservation and momentum's flux are theorems on the lattice; the click as the local, whole measurement, with the record never ending; the families as bands of that rule; the cloud as the body of nature with its three masses one, and the compact pixel as collapsed matter; the forms of nature arrived at from the coefficients and set beside their known forms; and one experiment's row with its blind expectation and its result. Not claimed: that nature is so; a derivation of the Lorentz group, which is no symmetry of the lattice (the lattice's symmetries are the 48 signed permutations of the axes); Born's rule or Bell's correlation, which the local click does not give and which stand as findings against nature and as an open row; the fine-structure constant, the mass ratio of the proton to the electron or what fixes the Node clock,

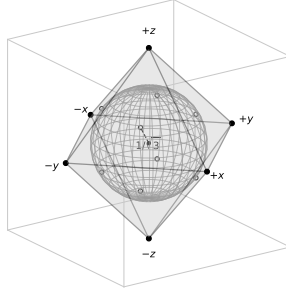


Fig. 2 The six Ports of one Node, $\pm x$, $\pm y$, $\pm z$, and the octahedron they span: the Nodes one interval away. The maps of the six Ports onto themselves that keep opposite Ports opposite are the 48 signed permutations of the axes, the symmetry group of the cube, under which the law is covariant. Drawn from the definitions; no run.

which are open by name (Section 8.1). The paper holds no run of its own; every number of a run is the documents' and is cited so.

2 The general formula

This section states the object and proves what is exact on it: the lattice and its families, the one line with its coefficients, its reversibility, what its acts can and cannot form, the paces with their guard, the conserved form, the tension as the wave's own stress, the one write, the interval and its inverse, and the three channels through which one family acts on another [1, the GameBoard and the families; Rule3; the primitives].

2.1 The objects

A Node is a physical location. The GameBoard is all the Nodes together, a box of (N_x, N_y, N_z) Nodes; each axis is periodic (the box a torus on that axis), open (a face beyond which no Node lies, read as 0) or folded (an axis of one layer, whose two Ports return the Node itself), as the world declares. Two Nodes that differ by one step along one axis are joined by a Link; each Node has six Links, one through each of its six Ports, $\pm x$, $\pm y$, $\pm z$ (Fig. 2). An inner face is the face's rule declared inside the board: a plane of Nodes across an axis, beyond the board but for the gaps the world declares; a Node beyond the board is read as 0 through every Port and reads 0 through its own, so no level, current or source crosses its Links, and no body, message or detector stands there. The board's shape is a declaration of the file, like the wraps and the detector Nodes, and no number of the engine. The local information of a Node is its NodeState: the levels and remainders of the families at that Node, and nothing else; no register, no table, no formula and no draw stand at a Node [1, the objects][3, nothing at a Node but the law's numbers].

A family is one kind of physical value with its own declarations: its pair [num, den] and what it holds. Its record is two levels per Node (now and before) with a remainder per Node; a family of quanta carries two such level pairs, the real one and its second, the rotation sense (Section 3.4), and a count with the count's remainder. A body is a set of Nodes on which a family's count stands, one connected region; a message is a record the world lays and no body, a packet of a family of quanta with its axis, its wave number, its amplitude and its envelope declared [1, the objects].

2.2 The line

At every Node, each interval, every family steps by the first line of Eq. (1): the six arrivals, each the neighbour's level through its Port (0 beyond an open face, the Node's own level on a folded axis), the Node's own level now, minus its level before, plus the remainder kept at the Node, divided once by the wall w ; the quotient is the next level and the remainder stays. The remainder is the Node's own and never travels [1, the line].

At the vacuum's pace $p_0 = p_a = \Gamma$ the self coefficient vanishes and the line is the plain second-order wave rule of the pair,

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

whose rotation at the wave vector $\mathbf{k}$ is $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$: at wave number zero $\cos \omega_0 = \text{num}/\text{den}$, and for the massless pair [1, 1] along one axis $\cos \omega = (2 + \cos k)/3$, so that $\omega = |k|/\sqrt{3}$ as $k \rightarrow 0$. Light's speed at long wavelength is therefore $1/\sqrt{3}$ Links per interval, under the causal bound of one Link per interval, the arrival's own speed; the two differ, and what in nature is one Link and one interval rests on the units of Section 8 [1, the lattice constants].

The clock once and the Link twice. From the coefficients, the rotation at a Node of the clock's pace p_0 and the Link's paces p_a is (DERIVED)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{ den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a), \quad (3)$$

the mass term scaling with the clock's square and the band term with the Link's pace. Two readings follow and no third [1, the clock once and the Link twice]. (i) *The clocks shift alike*: at $k = 0$ every massive band's rest rotation obeys $1 - \cos \omega_b = (p_0/\Gamma)^2 (1 - \cos \omega_0)$, one factor for every family, so two bound bodies at one Node shift their clocks alike, and a packet's fall in a pace gradient does not depend on its family beyond its band's own shape. (ii) *The Link is twice*: along a Link every wave's speed falls with $p_a = \Gamma - 2c - t_a$, the held level c entering the Link twice and the clock once, so light, which reads the held rows as every family does, is delayed and bent by twice the clock's share (Section 6). The line is not conformal: a light wave at a fixed wave number scales with $(p_a/\Gamma)^2$ and a bound mode at rest with $(p_0/\Gamma)^2$, and the two differ already at first order in c/Γ .

A held family's field steps at its row's pair with the pace 1 and the wall 3 den, at first order: for the massless row $a_{\text{next}} = (S_6(a_{\text{now}}) - 3 a_{\text{before}} + r) \text{ div } 3$, S_6 the sum over the six neighbours, and for a row with a gap $3 \text{ den } a_{\text{next}} + r' = \text{num } S_6(a_{\text{now}}) - 3 \text{ den } a_{\text{before}} + r$, the same call at the row's own pair [1, the vacuum's row steps at the divisor 1].

2.3 The direction

Let σ be the direction, +1 forward and -1 backward, let div and mod be the floor division and its remainder in $[0, w)$, and let ρ be the remainder carried in. With

$$u = \sigma \left(\sum_a R_a \text{arr}_a + S x - w y \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (4)$$

forward $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and backward $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Theorem 1 (the direction). *The backward call returns (a_{before}, r) bit for bit; the reversibility of every step needs no second function.*

Proof. Forward, $\sigma = 1$, the line is the definition of div and mod . Backward, $\sigma = -1$: write Y for the right side's first two terms, $\sum_a R_a \text{arr}_a + S a_{\text{now}}$. Then $u = w a_{\text{next}} - Y + r'$ and $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$. From the forward line, $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$ with $0 \leq r < w$, whose ceiling divided by w is a_{before} exactly; and $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$. Checked bit for bit on 20,000 random cases against a forward and a backward function written separately [1, the direction]. $\square$

2.4 The four acts and what they can form

Rule3's line is one act; declared coefficients make it four: (a) the read, the line over the GameBoard with its six arrivals; (b) the one-Node step, the line with the neighbours' reads declared 0, $a_{\text{next}} + a_{\text{before}} = (a/b) a_{\text{now}}$, the rotation by the angle whose doubled cosine is a/b (Chebyshev's recurrence), which with $a/b = 1$ and the coefficient on a_{before} declared 0 keeps a level and with $a/b = 2$ from $(0, 1)$ counts; (c) the division, the line with $R = S = 0$, Euclid's division with the remainder kept; (d) the load, a declared integer in the numerator, how a count enters a level. A composition of the acts is a finite sequence of them, each with its coefficients declared in the files, applied to declared levels [1, the four acts].

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

Proof. Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of such functions are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product $\text{now}_i \text{before}_j$ in the pair $(\text{now}_i, \text{before}_j)$ is 1. So nothing bilinear comes from the acts alone. $\square$

What the theorem separates: every product of two levels in the engine is a *booking*, a reading of a bilinear form with declared coefficients and no write (Section 3.1); the one thing written from a product is the count, by the count's line, and that product is Rule3's own conserved current (Section 2.6).

The three tests and the six verbs. A line enters the law only if it is generic (one primitive with declared integers, no family's name and no kind), vector (one of the acts on the state, no root and no float) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers); a hypothesis that needs more is stated under its own name, outside the law. Every integer of the law is made of six operations only: the translation of an accumulator by its rate, the bilinear form with a declared matrix, the group-ring addition, the permutation, the evaluation at the primitive root of unity, and the Euclidean division with the remainder kept; a root, a true division or a draw is outside the law. No act of the interval takes a square root: the integer square root stays only at load and in the generator, as the fixed point of the division act iterated, $x \leftarrow (x + n \text{ div } x) \text{ div } 2$ from above until it repeats [1, the three tests of every rule][3, the six verbs; the root leaves everywhere].

2.5 The paces and the guard

The pace of a Node for a family is what its reads make of the Node clock:

$$p_0^2 = (\Gamma - c)^2 + c^2, \quad p_a = \Gamma - 2c - t_a, \quad c = \sum_{\text{reads}} \text{weight} \times \text{by} \times \text{the read family's time level}, \quad (5)$$

t_a the axis content from the read families' tensions on the axis a (each weight $\times$ by $\times$ the aa part $\div 2$, one division per read per axis, rounded at the read since the pace is a coefficient and no level). A positive weight is a hollow (the read family's level slows the clock, an attraction); a negative weight is a hill. By sign the weight is multiplied by the reader's q at the Node, the sign of its own rotation sense there (Section 3.4). A family with no reads steps at $p_0 = p_a = \Gamma$, the plain rule. The clock's slowing is this pace: a record at a Node of pace p rotates and moves as a record at Γ does, with its intervals p/Γ as long. Every family of quanta reads every holder of the content at the weight 1, a row without a gap and a row with a gap alike as its stepped field, $x = (L_b + L_g)/\Gamma$ with L_b the binding holder's level and L_g gravity's, and every holder of the sign by its own q ; a family sources what it reads at the weight it reads with [1, the paces; the reads are the held row's pair's].

The clock's second order is the composition. Each unit of the level slows the clock as it already stands, $p_0 = \Gamma(1 - 1/\Gamma)^c$, whose square to the second order is $(\Gamma - c)^2 + c^2 = \Gamma^2(1 - 2\Phi + 2\Phi^2)$ in the level's share $\Phi = c/\Gamma$; Rule3's coefficients take the paces' squares as written, with no division act, no rounding and no root, and $p_0 = \Gamma - c + c^2 \div 2\Gamma$ to the unit is a reading of the algebra; the Link's pace stays $\Gamma - 2c - t_a$, the level entering it twice [1, the clock's second order is the composition]. Section 6 reads the perihelion off this second order.

The guard, on squares, once at load. $0 < p_a$ and $p^2(\text{den} + \text{num}) \leq 2 \text{den} \Gamma^2$ at every Node, with the edge's square $P^2 = 2 \text{den} \Gamma^2 \div (\text{den} + \text{num})$, Γ^2 for light and above it where $\text{num} < \text{den}$, read on the clock's square and on each Link's pace alike over the whole initial state and never in the interval. Below 0 a pace is no clock; beyond P the step is unstable: the mode at wave number π on every axis has the factor $2 - 2(1 - \text{num}/\text{den})(p_0/\Gamma)^2 - 4(\text{num}/\text{den})(p_a/\Gamma)^2$, which is $-2 \text{num}/\text{den}$ in the vacuum and stays at or above -2 wherever p_0 and p_a are both at or below P , since $(P/\Gamma)^2(2 - 2\text{num}/\text{den} + 4\text{num}/\text{den}) \leq 4$ exactly when $P^2(\text{den} + \text{num}) \leq 2 \text{den} \Gamma^2$ (THEOREM). In the run no act reads a Node to refuse: every coefficient is a square, $R_a = 2 \text{num} p_a^2$ passes through 0 as the content reaches $\Gamma \div 2$ (the Link closes: the horizon) and reopens mirrored beyond it, and beyond P the mode at π grows until the host's trap on the arithmetic's width ends the run; the report gives the pace's minimum from the final state as a diagnostic [1, the guard][3, no one asks a Node; the root leaves the run].

The band at a pace. A reading of the line and no new rule: at a Node of paces p_0 and p_a a record's rotation at the wave number k along an axis is Eq. (3), so the band of rotations a body's Nodes carry narrows with the paces; a record whose rotation lies outside the band there is evanescent inside and reflected whole, and one whose rotation lies inside is slowed to the band's group velocity. Light, [1, 1], meets a total mirror where $p_a < \Gamma \sin(k/2)$, and a window otherwise [1, the band at a pace]. At a Node of content $x = L/\Gamma$ light's line has $S/w = 2 - 2(1 - 2x)^2$ and $R_a/w = (1 - 2x)^2/3$, its band is $[2 - 4(1 - 2x)^2, 2]$, and at long wavelength $\omega = (1 - 2x) |k|/\sqrt{3}$: light's speed is $c_0(1 - 2x)$, an index $n = 1/(1 - 2x)$, $n - 1 = 2x$, the content read twice (Fig. 3) [1, the rows against nature, (b)]. At $x = 0$ light's zone-corner mode sits at -2 with no margin, and at any negative content it grows, $\cosh \mu = 2(1 - 2x)^2 - 1$ with μ the growth per interval; the vacuum content of Section 4.1 gives it the margin $8 x_{\text{vac}}$ [1, what is open, item 22].

The lattice constants. The six Ports' own, exact and no number of nature: the massless row's rest around one static source of s quanta per interval at the divisor 1, by the start's iteration on an open box with its faces' constant image removed, is $3 G(r) s$ with $3 G(0) = 0.7582$ at the Node and 0.2582, 0.1286, 0.0826, 0.0608 and 0.0483 at 1 to 5 Links, toward $3/(4\pi) = 0.2387$ far away (COMPUTED); of which 0.76 s at the Node and 0.24 s/r beyond are the continuum's; and light's $1/\sqrt{3}$ [1, the lattice constants].

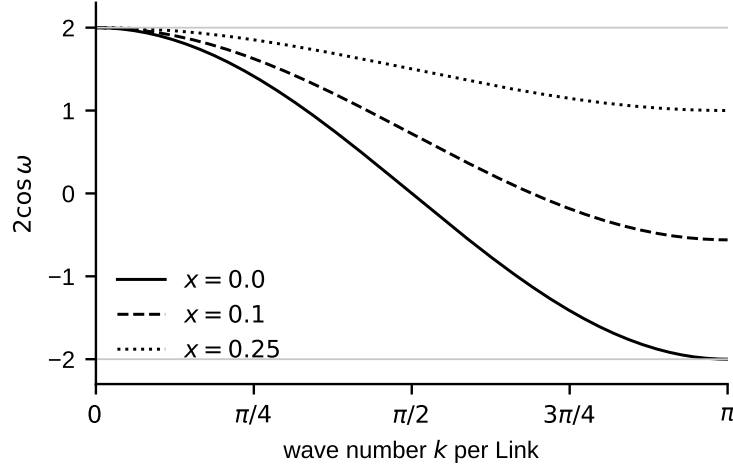


Fig. 3 Light's band at a content $x = L/\Gamma$ read into the paces (DERIVED, Eq. (3) at num = den): the doubled cosine of the rotation along one axis against the wave number, $2 \cos \omega = 2 - 2(1 - 2x)^2(1 - \cos k)$, at $x = 0, 0.1$ and 0.25 . The band's bottom rises with the content, its top stays at 2; at the horizon $x = 1/2$ the band closes to a point. The slope at $k = 0$ is light's speed, $(1 - 2x)/\sqrt{3}$ Links per interval. Drawn from the definitions; no run.

2.6 The conserved form

The read matrix $2 \text{ num } p_i^2$ is not symmetric where the paces differ, so the form carries the weight $1/p_i^2$ on the Node terms and the plain current on the Links:

$$E = \sum_i \frac{w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i}{p_i^2} - 2 \text{ num } \sum_i \text{ now}_i S_6(\text{before})_i. \quad (6)$$

Theorem 3 (the conserved form). *At constant paces E is exact under the line without the rounding; with the rounding each Node's remainder adds at most one unit of the level per interval to the walk of E . Its Link term, the plain current $\text{num}(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$ through the Link ij , is the current of Rule3's own conservation law.*

Proof. Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ with $\mathbf{M}$ the matrix whose diagonal is S_i/w and whose entries on the six Links are R/w . Let $\mathbf{D}$ be the diagonal matrix of the weights $1/p_i^2$; then $\mathbf{DM}$ is symmetric, since $R_a = 2 \text{ num } p_a^2$ at the reading Node and the weight $1/p_i^2$ cancels it on both sides of every Link. For a symmetric $\mathbf{K} = \mathbf{DM}$ the quantity $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D} \mathbf{x} + \mathbf{y} \cdot \mathbf{D} \mathbf{y} - \mathbf{x} \cdot \mathbf{K} \mathbf{y}$ satisfies $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$: with $\mathbf{a}_{\text{next}} = \mathbf{M} \mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$, $\mathbf{a}_{\text{next}} \cdot \mathbf{D} \mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K} \mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D} \mathbf{M} \mathbf{a}_{\text{now}}$, which is the other side. Multiplied by w this is E up to the Link term's sign convention [1, the conserved form]. $\square$

E is a GameBoard reading. It changes by the work term where the paces move: the total form $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$ of a record under paces that change in time obeys exactly $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}} \rangle$ (THEOREM, by substitution), 0 at fixed paces and negative where a well deepens; the count's line, moved by the flux alone, keeps the count meanwhile: the mass defect is the form's fall at a kept count (Section 5) [1, the mass defect is the form's fall at a kept count].

2.7 The tension: the momentum's flux is the wave's own stress

The momentum density along the axis a at the Node i is the count's line's current per axis, $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$, and Rule3's line gives exactly, at constant paces (THEOREM),

$$w [P_a(t+1) - P_a(t)] = \sum_b R_b [G_{ab}(i-b) - G_{ab}(i)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

$G_{ab}(i)$ the stress booking on the b -Link $(i, i+b)$ at one time, the levels now alone; the proof substitutes $w \text{ now}' = \sum_b R_b (\text{now}_{i+b} + \text{now}_{i-b}) + S \text{ now}_i - w \text{ before}_i$ into $w P_a(t+1)$: the S term cancels, the w term is $w P_a(t)$, and the R term is the difference of the fluxes. So the flux of the a -momentum across the b -Link is $-(R_b/w) G_{ab}$, the second booking of the conserved form (its Link term shifted by one Link), and the tension on the axis a at the Node, the part Rule3 reads into the Link's pace and the source of the held row's axis part aa , is the momentum's flux, the mean over the Node's two a -Links,

$$T_{aa}(i) = -\text{num} (G_{aa}(i) + G_{aa}(i-a)) \text{ div } 2, \quad (8)$$

$$G_{aa}(i) = \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}),$$

the neighbour's own difference brought through the Port and no other shift: the tension on a Link is the booking of its two ends' own readings, a level and a difference each, and a Node sources its tensions with half of each of its Links, so the line is local as F is. With the rounding the identity holds up to the remainders' term, $-r'_i \Delta_a \text{now}_i + \text{now}_i (r'_{i+a} - r'_{i-a})$, exactly. It is read and never written, quadratic in the amplitude as the count is: a plane wave of amplitude b at the wave number k has $G_{ab} = -2b^2 \sin k_a \sin k_b$ exactly, so the flux along the wave is positive, as a wave's pressure is; a uniform record gives 0; a standing body's tail carries a tension that does not vanish in the mean; for a packet of count c and velocity $\mathbf{v}$ it is $c v_a v_b$ within the rounding, the dust's stress with no count in any divisor and nothing undefined at a count 0. With the sign as written a wave's stress deepens the content around it: light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification][3, the tensor part is the wave's own stress].

2.8 The one write

Every write into a family's level on the GameBoard is one act, the third line of Eq. (1): $(w q(\text{Node}) + r) \text{ div } E$ at a Node each interval, the division act with the remainder r carried at that Node; q the writer's own quantity there (a family of quanta's well $D_i \text{ div } T$, its Wronskian's quanta, or its tension T_{aa}), w the one weight of the pair of families (whoever reads with w sources with w), E the written row's divisor (E_s ; $E_s W_c$ for a tension, one act per axis). The hold, the well and the start are its instances; no number is the act's own. Generic (the rows' keys), vector (one carried division), local (the Node) [1, the primitives][3, the one algebra]. The form $D_i = \text{now}_i^2 - \text{next}_i \text{ before}_i$ of a record at a Node is its well's source: for a standing record it is $B^2 \sin^2 \omega_b$ at every interval, B its peak [1, the gate reads the form at any phase].

2.9 The interval and its inverse

The engine steps every family from the state at the interval's start to the state at its end; every read is of the start's values, one Link per interval and nothing in zero time. Before the first interval the whole initial state is checked once against the guard; then the interval has four acts, in order:

(i) the signed read, every family's paces from the held families' levels at the start; (ii) Rule3 on every record, each family of quanta's two level pairs by its pair and its paces, each holder of the content's parts at its own pair at the pace 1; (iii) the count's line and the sense's line on every family's count and sense, the tension on each axis booked beside them, and the clicks at the Nodes declared to report; (iv) the hold, each held family's time part gaining its source over its divisor and its tensions the sources' momentum fluxes; light is born here, by the write of a rotating body's varying Wronskian into the sign holder's record [1, the interval][2, the interval].

Backward, the acts run by Rule3's direction -1 : a held row's hold back once every family that sources it is booked back, a family of quanta booked back once its own hold is off, so the holder of the sign goes before the rows it sources; every interval returns bit for bit, since every level a family read at the interval's start is one the state after the interval still holds, and only the lay of the first act is not taken back. The back-in-time gate is the whole run: N intervals forward and N back, every array of every family compared bit for bit at every interval on the way back, MATCH or MISS naming the first interval and the first Node that differ; a look is not run before it says MATCH on the look's own world (THEOREM by construction, from Theorem 1 and the count's line's inverse, Section 3.2). On the two slits' worlds of Section 7 it says MATCH over their 130 intervals [1, the interval; what is open, item 21][3, the board goes back in time, as a gate].

2.10 The three channels

One family acts on another in three ways and no fourth (Fig. 4): *the read*, at (i), a held row's level times the read's weight into the reader's paces, the clock once and the Link twice, multiplicative, the whole of gravity, of binding and of the charge's pull; *the write*, at (iv), the writer's own quantity through the one weight and the written row's divisor into the held row's level, additive, the source of every field and the birth of light; and *the click*, at (iii), the count's line's move of one whole quantum, the measurement alone. No product of two levels is written; only the current is read and the count written from it. No bilinear coupling between families is in the law: the world of clicks (bodies, detectors, the counts) and the world of Nodes (the records) run as one loop and touch only at the click and at a body's Nodes; a record sources the families its own family reads, so every pair of families that read each other is reciprocal, action and reaction one key [3, the coupling is the click alone; light is born by the write].

3 The click

The click is one mechanism: the count's line carries every family's quanta, and a Node declared to report reads each quantum that arrives whole [1, the primitives]. This section states the booking, the count's line with its theorem, what the click ends (nothing), the sign, the birth of light, absorption, the price of locality and the postulates.

3.1 The booking

The current of a record into the Node i from its neighbour j through their Link over an interval is the bilinear form of its two levels at the two ends, $F_{ij} = \text{num}(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$, positive inward: a record of wave number k travelling from i to j has $F_{ij} = -2 \text{ num} |\psi|^2 \sin \omega \sin k$, below 0. It is antisymmetric, $F_{ji} = -F_{ij}$, and it is the Link term of the conserved form (Theorem 3). The booking reads it at a Port and writes nothing [1, the booking].

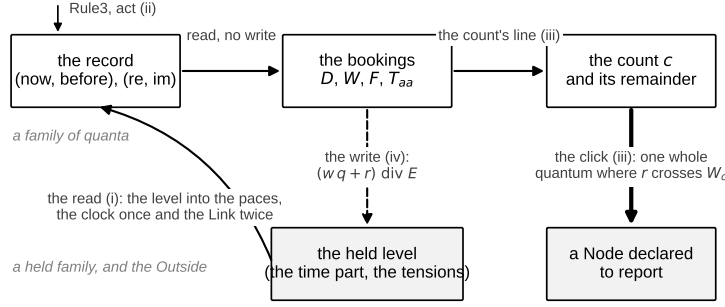


Fig. 4 The three channels through which a family acts on another, drawn from the interval's four acts (Section 2.9). A family of quanta steps its record by Rule3 and books its form and its current; a held family carries a level. The read (i) takes the held level into the record's paces; the write (iv) takes the record's form, its Wronskian's quanta and its tension into the held level over the row's divisor; the click (iii) moves one whole quantum of the count where the remainder crosses the wall, and a Node declared to report reports it. Every arrow is one line of Eq. (1). Drawn from the definitions; no run.

3.2 The count's line

Every family of quanta carries a count c , the number of its quanta at a Node, one count per family over the GameBoard; its wall is $W_c = 3 \text{ den } T$, T the quantum's action; its read is the current. At every Node the second line of Eq. (1) holds: Rule3 with the coefficient 1 on each axis's inflow through its two Ports, W_c on the count and the wall W_c . The click is the division: a quantum moves whole from one Node to the next exactly when the remainder crosses W_c , and never a fraction of one. The inverse is the same line with the current reversed, $W_c c_{\text{now}} + r = W_c c_{\text{next}} + r' - \sum_j F_{ij}$, exact, the currents read from the interval's levels as the forward step read them. The line has no other act: a total below zero gives the count -1 with the remainder in range, the hole, which the line conserves and fills; a clamp at 0 would create quanta, and a block of a Node's currents would be a branch on a Node's value that loses the inverse, so neither is a rule [1, the count's line; the inverse][3, nature pushes toward whole numbers].

Theorem 4 (the count is conserved). *Over any region of Nodes, $\sum (W_c c_i + r_i)$ changes in one interval only by the currents through the region's boundary Links.*

Proof. Summing the line over the region, each interior Link's current enters one Node's right side as $+F$ and its neighbour's as $-F$ and cancels; what remains is the sum over the boundary Links, a telescoping sum, exact in integers. On a periodic GameBoard with no face the sum is constant to the bit [1, the count's line]. $\square$

The remainder's origin, and the lay. The remainder at a Node starts at the origin $W_c \text{ div } 2$: started at 0, a Node with no quantum reads the count -1 at its first outward swing. The count is laid once, at the first act, from the weighted share, the conserved form's Node term over the Link's pace squared less the plain Link term, $e_i = [w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i] \text{div } (2p_i^2) - \text{num now}_i S_6(\text{before})_i$ with p_i the Link's pace at the Node, in the current's units, $W_c c + r = e_i + W_c \text{ div } 2$; its change over one interval of Rule3 is exactly $\sum_j F_{ij}$ at every Node, in the vacuum and in a well alike, so the count's line is the continuity of Rule3's conserved form everywhere

(THEOREM). After the lay the count moves by the line's flux alone. In the vacuum the share is the plain share, whose sum over a closed GameBoard is 3 den $\sum_i D_i$; in a well the plain share is the local form D that sources the fields and no count, so a body in its well is lighter in every field it sources than in quanta [1, the remainder's origin; the lay and the wall][3, the lay from the weighted share].

The count's unit. The engine's count, the weighted share at the Node's own paces, is the count the law conserves; a count in the vacuum's share (the plain share at the vacuum's coefficients) counts the same record less at a deep Node, 3.7 times less at the Node of the 2,000 pixel of Section 5 and 3.1 times less in all, and the same to a percent at a cloud, whose content is 0.005. So a world is declared at the engine's count, the unit-free numbers stand, and the counts of the compact branch are restated by the generator's conversion at the Node's paces [1, the count's unit].

The velocity. The record's wave number is conserved by the record's line where no pace gradient stands (Rule3 is translation-invariant), and the count's line is the continuity equation of the current in integers, so the count's centroid moves at the current's group velocity, with no leak beyond the remainder; a body at rest has zero net current at every Node in the mean and its count stays in place; a moving body's clicks keep its velocity [1, the velocity].

3.3 The click ends nothing

Assumption 1 (the click ends nothing). The click is the count's line's move of one whole quantum where the remainder crosses the wall, at one Link, local; the record goes on after it and nothing ends anywhere. The count that arrives at a detector's Node is the detector's click, a detector's count is a family's count at its Nodes, and a detector reports and takes nothing. The ending of a record at a click by the detector, a Node that takes the quantum into its own count, is decided against and is no hypothesis of the law.

This is the owner's decision of 2026-09-30 [3, the click ends nothing], and it is the law's claim to be tested against nature, not a deficiency: perhaps nature is local. What follows from it needs no further act. One quantum is reported once and never twice, by the count's conservation (Theorem 4) and by no act at a distance; no signalling holds trivially, since nothing at one detector changes anything at another; the Outside reads the clicks alone; and the click is the one reading at a body's scale. A click has a family and an axis: the family whose count moved and the Port the quantum crossed, its axis and side, the count's line's tally; the three tallies are what a detector reads of the click's axis, no charge of any body and no force of their own [1, the count's line; the third and the colours, closed as absent].

3.4 The sign is the rotation sense

A family of quanta carries its record as two level pairs, (re,im), two solutions of the same line stepped by one call of Rule3 with one set of coefficients, each with its own remainder; a held family's parts stay real. The charge density at a Node is the booking

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (9)$$

the Wronskian of the two pairs, which Rule3 conserves exactly with the conserved form's weights $1/p_i^2$ (THEOREM, **DM** symmetric), its local change the Link current num $(\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$ at the levels the step started from; the sense is laid once at the first act from $w W_i \text{div} (2p_i^2)$ at the count's wall and moved by the count's line on that current, $\sum (W_c k + r)$ conserved to the bit. A real record ($\text{im} = 0$) has $W = 0$, so light, the sign holder's own real record, is neutral; a rotating record has W

of either sign by its sense, $-A^2 \sin \omega$ for $e^{+i\omega t}$ and $+A^2 \sin \omega$ for $e^{-i\omega t}$. The reader's q at a Node is the sign of its own sense there; the holder of the sign is sourced by the Wronskian's quanta, $(W_i + r) \div T$ each interval, over its divisor; pair creation is the conversion between the two senses; the coupling goes as the sign, never as Q/M , so an electron and a proton accelerate alike in one field [1, the sign is the rotation sense].

The charge is the record's, not the count's. For a record $(\text{re}, \text{im}) = (A \cos \omega t, eA \sin \omega t)$ the booking is $W = -eA^2 \sin \omega$: the charge is the count's size times the ellipticity e of the rotation, a property of the record that Rule3 conserves (the two levels obey one linear line, so their ratio is carried, not written). A circular quantum ($e = 1$) has $|W|/D = 1/\sin \omega$; a nearly real cloud of 1,836 quanta with $e = 1/1,836$ has the charge of one circular quantum; a body is one mode with one sense and one ellipticity at all its Nodes. What it costs, a finding against nature: nothing in the law quantises e , so a body's charge is continuous, and $|q_p| = |q_e|$, nature's to 10^{-21} , is an input and not derived; the line that would derive it is a hypothesis under its own name, *the second level's amplitude is a count of its own* (Section 8.1) [1, the sign is the rotation sense; the hypotheses under their own names].

3.5 Light is born by the write

Light is the sign holder's own record: the holder of the sign is one record whose time part is its real level pair, stepped by Rule3 at the paces of its reads (the content, so light is bent) and written by the hold from the Wronskian's quanta of the families that read it by sign. A static rotating body writes a static level (its well in the sign holder, Coulomb's) and radiates nothing; a breathing or moving body writes a varying source at its Nodes, the row carries it away at one Link per interval in the vacuum, and that wave is the light (DERIVED). Its count needs no lay and no conversion: every family runs the count's line on its own record, a held family's count starting at 0 everywhere where its record is 0 and moved by its currents, so its total stays 0 within the remainders' walk, quanta and holes -1 together, the line's own pair creation: a photon born with its hole. A body's quanta that leave it stay its family, messages of matter, and the matter count is conserved by the line to the bit; no quantum changes family at a crossing, no ledger of bodies stands beside the Nodes, and a body's Nodes are where its family's count stands about its declared Nodes, derived when a report needs them and never kept. A body reads the sign holder by its own sense at its own Nodes, so a breathing body reads the light it writes and its pace answers, the radiation reaction, in the same loop [1, light is born by the write][3, light is born by the write; the coupling is the click alone].

3.6 Absorption is the read by the sense and the write

Assumption 2 (absorption). A family of quanta reads the sign holder's level by its sense, so the light's level at a body's Nodes enters the body's paces, a drive at the light's frequency, resonant where it meets the body's rotation (the parametric line, at twice the rotation for a modulated pace), and the body's write answers, the driven body writing light of its own; behind the body the written wave and the incoming one cancel where they are out of phase; where the wave is cancelled its current stops, so the light's count that arrived at the body's Nodes does not leave, the count's line piles it there as light's count with no wave to carry it, and a Node told to report reports the rise. A light quantum never becomes a matter quantum; the conversion act is refused as unnecessary and as a branch, and matter's count stays the baryon number.

This is the owner's decision of the same day on the advisor's line [3, the click ends nothing][1, the count's line]; it is a body's act and no new act of the law: the read (i) and the write (iv) with

the sense in them, and the count's line as it is. So feeble light and a lone quantum spread over many Nodes never click at a bare detector Node (they move no remainder over W_c there), and at a resonant absorber one quantum clicks once, locally: a body that absorbs the whole wave over its resonance time integrates the wave's current into its Nodes' remainders, whose sum is the wave's count, one quantum, crossing once. Nature's detector is a body and not a Node. The free electron's detection is the same with matter's line, and the radiation reaction is the same loop. An absorber in one slit takes that slit's wave and the screen shows the single slit's pattern, nature's loss of the fringes by a detector in the path with no act at a distance; a bare detector Node in a slit changes nothing, and a marker that does not absorb has no line. What the absorption look reads, computed with the read alone before the write is built, is Section 7.1.

3.7 The price of locality

The law's word against nature, decided and against nature until tested [1, what is open, item 4]:

- (i) *The single-quantum statistics.* A wave split among many absorbers gives each a fraction, and at most one crosses (the one over a half) or none; no statistics are in it, and Born's rule is not in it. Nature builds the fringes click by click from a dilute beam, at places weighted by the wave's intensity.
- (ii) *The bare detector Node.* A lone quantum spread over many Nodes never clicks at a bare detector Node, and a reporting detector Node in one gap of the two slits changes nothing of the screen's row (the run of Section 7 shows both); nature's which-way loss of the fringes needs an absorber, which the law has as a body (Assumption 2) and not as a Node.
- (iii) *Bell's row.* Two labels of one written wave, each clicking alone at its own end: the pair's birth, the turn at each end and the labels' clicks are open under the count's line; the blind expectation stands, $E(a, b) = \cos 2(a - b)$ with each side's marginal one half and $S = 2\sqrt{2}$ at the settings 0, $\pi/4$, $\pi/8$, $3\pi/8$, and nothing here derives it [1, the rows against nature, (h)].

The two slits' dilution series and the absorption look test (i) and (ii), and their result is the law's answer, not a line to add. The paper writes these findings as they stand.

3.8 The postulates

1. The world is Nodes and Events. Space is the GameBoard; an Event is a wall crossed by a remainder at a Node: a level stepped, a count moved, a click; nothing else happens.
2. A Node holds bounded local information: its NodeState, fixed in size, read from itself and its six neighbours; no read beyond them, no list that grows with the world, nothing of the past kept. A self-field estimated or subtracted from anything global is forbidden, whatever the size of its final answer.
3. Consistency is local and causal: an Event first changes its own Node and travels Link by Link, one Link per interval, each Node updating on receipt under the same rule; there is no exception, and no quantum changes family at a crossing.
4. The causal speed is one Link per interval, built in; every other speed is a rational, a count of Links over a count of intervals.
5. Every physical calculation is on bounded integers; there is no float, no root and no draw in the law; a Node keeps only the law's own numbers.
6. A measurement is a detector's click, an action of the law on the state: a count arrives at the detector's Node and the detector's count changes. Only a click is compared with nature or pinned as an expectation; displays and the host's readings read state and write nothing.

7. The Inside is the GameBoard, where no one measures; the Outside is the detectors and their clicks, the only thing claimed to represent nature.
[1, the postulates]

4 The families from the rule

This chapter is written in the second round; its content is the outline's: the pairs and the exact bands, the rule's own universe and its five rows, the two rows (the binding holder and gravity under a divisor, decided with the caveat), the vacuum content and the count's unit [1, a family's declaration].

4.1 The vacuum content

Written in the second round: the massless row rests at a declared level c_{vac} everywhere, the law's cosmological constant, positive as nature's [1, what is open, item 22].

5 The bodies

This chapter is written in the second round: what a body is, the two branches (the cloud as the body of nature, the compact pixel as collapsed matter), the three masses and the stable body theorem [1, the stable body].

6 The forms of nature the line arrives at

This chapter is written in the second round: the clock once and the Link twice, the redshift, the bending, the delay, the perihelion and the moving clock, each set beside its known form and none an input [1, the rows against nature].

7 The two slits

This chapter is written in the third round: the world, the blind row, the result and the findings against nature [1, the rows against nature, (g)].

7.1 The looks to come, blind

Written in the third round: the pixel look, the fall look and the absorption look, their blind rows as the documents hold them.

8 The constants, and what is open

This chapter is written in the fourth round [1, the constants; what is open].

8.1 What is open, and what is not claimed

Written in the fourth round.

9 The engine that computes it

Written in the fourth round, from the engine’s document [2].

Declarations

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Data and code availability. The law’s document, the engine that computes the rule and the figures’ scripts are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [4], under the MIT licence; the one run this paper cites, the two slits, is cited through the law’s document and names its commit there.

Author contributions. The author laid down every postulate and decision of the model, directed the work and wrote the paper.

The use of a large language model. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under his direction and instructions, for the software, the computations, the mathematical derivations and their checks, the physics of the law’s document and the drafting of the text; the model is not an author; the author reviewed and edited every statement and takes full responsibility for the content. No figure was made by a generative tool; every figure is drawn by a script from the definitions or from the documents’ rows.

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- [1] The law of the GameBoard, `docs/ALGEBRA.md` of the archive [4], cited by the name of its section: the GameBoard and the families (the objects, a family’s declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count’s line); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, the hypotheses under their own names, the postulates, the constants, the rows against nature); what is open.
- [2] The engine, `docs/ENGINE.md` of the archive [4]: the words, the NodeState, the interval, the loader and the files, the output, how to run a world, how to add a family, the gates.
- [3] The decisions, `docs/HIGHLIGHTS.md` of the archive [4]: the model owner’s decisions in force, one line each, cited by the line’s name.
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